

Maintenance Manual — Tide Motors Assembly Plant 7

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Plant	Tide Motors Assembly Plant 7	Effective Date	15 Jan 2026
Document Owner	F. DeLuca, Maintenance Engineering Manager	Classification	Controlled — Site Use

1. Purpose & Scope

This manual governs preventive and predictive maintenance of stamping presses, weld robots, paint robots, general assembly conveyors, and battery pack assembly tooling at Tide Motors Assembly Plant 7. Activities are aligned with the IATF 16949 quality system and TPM (Total Productive Maintenance) program.

2. Critical Asset Register

The table below lists the critical assets covered by this manual. Each asset has a detailed task package in the CMMS (SAP-PM / Maximo) linked to its asset ID.

Asset ID	Description	Maintenance Strategy
STP-2500T	Stamping press 2500 ton	Daily ram pad inspection; quarterly clutch/brake monitor.
RBT-WLD-cluster	30x weld robots BIW Cell 3	Tip dressing every 250 cycles; servo motor health monthly.
RBT-PNT-cluster	Paint robots in E-coat & topcoat	Bell cup cleaning per shift; atomizer overhaul every 2000 h.
BAT-FIX-07	Battery pack assembly fixture	Torque transducer calibration weekly; vision check daily.
CNV-MAIN-01	Main assembly conveyor	Chain wear monitoring; drive lubricant change quarterly.

3. PM Schedule Summary

Frequency	Activity
Per shift	Operator AM (Autonomous Maintenance) checklist; weld tip wear inspection.
Daily	Battery torque tool verification with master joint.
Weekly	Servo health analytics review; conveyor chain elongation measurement.
Monthly	Predictive analytics review per OEE bad-actor list.
Annual	Stamping press clutch/brake monitor certification; robot mastering.

4. Work Management

All maintenance work is authorized through a CMMS work order. Work orders are categorized as PM, PdM, Corrective, Emergency, or Project. Maintenance KPIs (Schedule Compliance ≥85%, PM Compliance ≥95%, MTBF) are reported weekly to the Plant Manager.

5. Procedures

5.1 TPM and OEE

Equipment effectiveness target $\geq 85\%$ per IATF 16949. Maintenance schedules are dynamically tuned by the Reliability team using time-series data from the MES.

5.2 Battery Tool Calibration

Critical fasteners on the battery line follow $Cm/Cmk \geq 1.67$ per the SPC plan. Out-of-control fasteners halt the line and trigger a containment per QC-TM7-CONT-04.

5.3 Robot Mastering

Robots re-mastered annually or after any collision. Mastering data archived in the robotics historian for 7 years per quality records retention policy.

6. References

- IATF 16949 Section 8.5.1.5 (TPM)
- ANSI B11.1 — Mechanical Power Presses
- Internal: TPM-TM7-PROG-01, QC-TM7-CONT-04

7. Revision History

Rev	Date	Author	Summary
1	04 Jun 2023	F. DeLuca	Initial issue.
2	21 Nov 2024	F. DeLuca	Added predictive analytics integration.
3	09 Jul 2025	F. DeLuca	Updated asset register.
4	15 Jan 2026	F. DeLuca	Refreshed PM frequencies and KPIs.

Approvals

Role	Name	Signature	Date
Maintenance Engineering	F. DeLuca, Maintenance Engineering Manager	_____	15 Jan 2026
Reliability Engineering	Sr. Reliability Engineer	_____	15 Jan 2026
Operations / Site Lead	E. Brennan, Plant Director	_____	15 Jan 2026

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